

Udit Kumar

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PROFESSIONAL OVERVIEW

Detail-oriented and motivated Computer Science Engineer with hands-on experience in cybersecurity, cryptography, network security, and software development. Skilled in Java and Python, with practical experience building AI-powered intrusion detection systems and post-quantum secure file transfer applications. Strong problem-solving abilities, quick learning mindset, and a keen interest in threat analysis, security research, and developing secure and reliable systems.

PROJECTS

CyberSentinel — AI-Powered Intrusion Detection System JUN 2026 – AUG 2026

- Developed an ML-based IDS using Random Forest to detect Normal, DoS, Probe, R2L, and U2R traffic.
- Trained on 125,973 NSL-KDD records, achieving 99.84% training accuracy.
- Integrated Scapy for packet analysis with a FastAPI-based prediction system.
- Deployed a web interface displaying attack types, confidence scores, and risk levels.

Post-Quantum Secure File Transfer Java, Kyber, ECDH, HKDF, AES-256-GCM, Dilithium FEB 2026 – MAR 20216

- Developed a hybrid secure file-transfer application combining Kyber post-quantum key encapsulation with ECDH for hybrid key establishment.
- Implemented HKDF-based key derivation to generate secure session keys and AES-256-GCM for authenticated file encryption.
- Integrated Dilithium digital signatures along with timestamp and nonce validation to provide authentication and protection against replay attacks.
- Built the application using Java 17, Bouncy Castle, and JavaFX, with successful file transfer, decryption, and integrity verification.

Team Task Manager — Full-Stack Project Management System React, Node.js, Express, MongoDB, JWT, Tailwind CSS MAR 2026 – MAY 2026

- Developed a full-stack task management application for project, task, and team management.
- Implemented JWT authentication and role-based access control for Admin and Member users.
- Built features for project creation, task assignment, status tracking, and team member management.
- Deployed the application with a responsive React frontend and Node.js/Express backend connected to MongoDB.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering | Chandigarh University, Mohali, Punjab 2026
Class XII | Shiv Devi Saraswati Vidhya Mandir, Budaun, Uttar Pradesh 2020
Class X | Shiv Devi Saraswati Vidhya Mandir, Budaun, Uttar Pradesh 2018

SKILLS

Java	Network Security	Teamwork
Python	Post-Quantum Cryptography	Adaptability
JavaScript	Machine Learning	Time Management